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InsertAnything

A Single Simulation-Trained Policy Achieves 95% Success Across Eight Real-World Precision Insertion Tasks at 0.02 mm Clearance

By [Authors, arXiv:2609.24511]

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InsertAnything trains a reinforcement-learning insertion policy entirely in simulation using only target poses and compact 3-axis fingertip forces, then deploys it directly to real robots—achieving 95.0% success across eight unseen geometries at sub-millimeter clearances and the first perfect score on ManipulationNet’s peg-in-hole benchmark.

AT A GLANCE

Problem: Precision peg-in-hole insertion at sub-mm clearances requires geometry-specific demonstrations; policies do not transfer across shapes.

Why it matters: Industrial assembly requires a single robot to insert hundreds of distinct parts; retraining per geometry is impractical.

Method: RL in simulation with decoupled gated reward, 3D fingertip force feedback, force smoothing, and state-independent action standard deviations.

Result: 95.0% success across 8 real insertion tasks; 20/20 on ManipulationNet; min clearance 0.02 mm.

Evidence: Cross-geometry and cross-clearance evaluation; 4-axis ablation study confirming each design choice.

The Big Picture

Precision insertion—placing a peg or bolt into a hole with clearances as small as 0.02 mm—is an archetypal industrial assembly skill. It requires tolerating position estimation errors, searching for alignment under contact, and recovering from jams. These challenges compound dramatically across geometries: the contact mechanics for a hexagonal peg differ from a round one, and the jam-recovery strategy that works for one fails for the other.

The standard robotics pipeline collects new demonstrations or builds a new controller for each geometry. InsertAnything takes the opposite approach: train a single RL policy in simulation on *one* geometry (hexagonal), using only a minimal sensor suite—target poses plus compact 3D fingertip forces—and show that the resulting policy generalizes to eight qualitatively different real geometries it has never seen.

The key insight is that the *skills* required for insertion—lateral alignment search, jam detection via force, and downward insertion progress once aligned—are geometry-agnostic. The policy must learn these skills, not geometry-specific trajectories.

Why Existing Approaches Fall Short

Imitation learning from demonstrations binds generalization to the demonstrated geometries; re-collection is required for each new part.

Model-based contact planning requires accurate part geometry and contact-mode enumeration, intractable for arbitrary shapes at runtime.

Vision-based RL adds perception errors on top of control errors and requires careful camera calibration; contact state is ambiguous from images.

Rich tactile sensors (GelSight, TacTip) provide contact images but introduce high-dimensional observation spaces and complex sim-to-real transfer for the sensor model itself.

InsertAnything uses the simplest possible contact signal—3D force vectors per fingertip—which is easy to simulate and calibrate to real hardware, while being sufficient to carry the alignment and jam-recovery information needed for precise insertion.

Core Mechanism

Observation Space. Two compact inputs:

1. A 6-DoF target pose for the end-effector
2. Smoothed 3D fingertip force readings $\tilde{\mathbf{f}}_t$

No camera input. The force signal provides the sole contact feedback.

Decoupled Gated Reward. The reward is gated by an alignment threshold ϵ_{align} :

$$r_t = \begin{cases} r_{\text{align}}(s_t) & \text{if lateral error} > \epsilon_{\text{align}} \\ r_{\text{insert}}(s_t) & \text{otherwise} \end{cases}$$

r_{align} rewards lateral error reduction; r_{insert} rewards downward depth progress. The gate prevents premature insertion reward before alignment, eliminating the jam-inducing failure mode.

Force-Signal Smoothing. A sliding-window average reduces noise without lag:

$$\tilde{\mathbf{f}}_t = \frac{1}{W} \sum_{w=0}^{W-1} \mathbf{f}_{t-w}$$

State-Independent Action Standard Deviations. The stochastic policy uses learned but state-independent

σ , preventing entropy collapse during the precision insertion phase and preserving exploratory behavior for jam recovery.

Technical Formulation

Policy:

$$\pi_{\theta} : (\mathbf{p}, \tilde{\mathbf{f}}) \mapsto \mathcal{N}(\mu_{\theta}(\mathbf{p}, \tilde{\mathbf{f}}), \text{diag}(\sigma^2))$$

where $\mathbf{p} \in \mathbb{R}^7$ is the target pose and $\tilde{\mathbf{f}} \in \mathbb{R}^{3n_f}$ the smoothed forces for n_f fingertips.

Alignment reward (lateral distance to hole axis):

$$r_{\text{align}} = -\sqrt{(x - x^*)^2 + (y - y^*)^2}$$

Insertion reward (depth progress):

$$r_{\text{insert}} = z^* - z_t$$

Full episode return:

$$R = \sum_t \gamma^t r_t + R_{\text{success}} \cdot \mathbf{1}[\text{depth} \geq d_{\text{target}}]$$

Learning or Inference Procedure

1. Randomise hole position (± 5 mm), peg orientation ($\pm 15^\circ$), friction in MuJoCo; train PPO on hexagonal insertion only.
2. Calibrate force scaling (simulation to real sensor).
3. Deploy on real robot—zero demonstrations, zero fine-tuning.
4. Evaluate on 8 novel real geometries (round, square, triangular, star-shaped, L-slot, and more).

What the Guarantee Says

The decoupled gated reward provides a structural inductive bias: the policy *cannot* earn insertion reward without first satisfying the alignment criterion. This eliminates the reward-hacking failure mode where policies achieve partial insertion depth by wedging the peg in a jammed position, which maximises depth reward but causes part damage.

Experimental Findings

- **Main result:** 95.0% average success over 8 real tasks; minimum clearance 0.02 mm
- **ManipulationNet:** First-ever 20/20 perfect score on the peg-in-hole benchmark under the Human-in-the-Loop protocol with fully autonomous insertion motions
- **Cross-clearance:** Policy trained at 0.1 mm generalises to 0.05 mm with modest success degradation

- **Force reduction:** Peak contact force $\approx 30\%$ lower than baseline without force feedback, reducing part wear

Ablations and Interpretation

No force feedback (pose-only): Success drops to $\approx 40\%$; the policy cannot recover from jams without force information.

No decoupled gate (linear sum reward): Success drops to $\approx 65\%$; premature insertion attempts cause hard jams.

State-dependent σ : Success drops to $\approx 70\%$; the policy becomes near-deterministic during insertion, losing jam-recovery exploration.

No force smoothing: Success drops to $\approx 80\%$; noisy force signals cause erratic recovery oscillations that destabilise alignment.

Reference

InsertAnything: Generalizable Contact-Rich Precision Insertion from Simulation to Reality.
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<https://arxiv.org/abs/2609.24511>